

SQL Standards & Core Concepts

Q. What is SQL? Discuss its features, the importance of SQL Standards, and categorize the different types of SQL commands with examples.

📌 **Definition to Remember**

SQL (Structured Query Language) is the standard **declarative language** for communicating with Relational Database Management Systems (RDBMS). It allows users to create, manipulate, manage, and query databases. SQL is governed by **ANSI** and **ISO** standards to ensure cross-platform compatibility.

1. SQL Standards

Standardizing bodies **ANSI** and **ISO** define the SQL standard to ensure compatibility across different database vendors (Oracle, MySQL, SQL Server).

Standard	Key Feature
SQL-86	First official standard
SQL-92	Introduced Joins and expanded query capabilities
SQL:1999 (SQL3)	Introduced object-relational features, Recursive queries, Triggers

Importance: Ensures core SQL commands work consistently across databases, **preventing vendor lock-in**.

2. Features of SQL

- **Declarative Language:** Users specify *what* data they want, not *how* to retrieve it.
- **Easy to Learn:** Syntax closely resembles natural English.
- **Highly Scalable:** Handles single-row queries to multi-million row analytics.
- **Comprehensive:** Covers data definition, manipulation, and access control.

3. Types of SQL Commands

Category	Full Form	Purpose	Key Commands
DDL	Data Definition Language	Define/modify database structure	CREATE , ALTER , DROP , TRUNCATE , RENAME
DML	Data Manipulation Language	Insert, update, delete, retrieve data	SELECT , INSERT , UPDATE , DELETE
DCL	Data Control Language	Manage user access & permissions	GRANT , REVOKE
TCL	Transaction Control Language	Manage transactions & data integrity	COMMIT , ROLLBACK , SAVEPOINT

Examples:

```
-- DDL
CREATE TABLE Student (ID INT, Name VARCHAR(50));
-- DML
INSERT INTO Student VALUES (1, 'John');
-- DCL
GRANT SELECT ON Student TO user_admin;
-- TCL
COMMIT;
```

★ Must-Write Points (for 10 marks)

1. SQL = Structured Query Language; standard language for RDBMS communication.
2. Governed by **ANSI** and **ISO** standards (SQL-86, SQL-92, SQL:1999).
3. SQL is **declarative** — specifies *what* to retrieve, not *how*.
4. **DDL**: structure operations (CREATE, ALTER, DROP, TRUNCATE, RENAME).
5. **DML**: data operations (SELECT, INSERT, UPDATE, DELETE).
6. **DCL**: access control (GRANT, REVOKE).
7. **TCL**: transaction management (COMMIT, ROLLBACK, SAVEPOINT).

⚡ Quick Recall

SQL → ANSI/ISO Standards (SQL-86, SQL-92, SQL:1999) → Declarative → DDL (structure) → DML (data) → DCL (access) → TCL (transactions)